

微积分甲 期中考试答案

1. 令 $a_n = \frac{1}{1+\sqrt{2}+\dots+\sqrt{n}}$ $b_n = \frac{1}{n\sqrt{n}}$... 3分

$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{1}{(\sum_{k=1}^n \frac{1}{\sqrt{k}})} = \frac{1}{\int_0^1 \sqrt{x} dx} = \frac{3}{2}$... 3分

$\therefore \sum \frac{1}{n\sqrt{n}}$ 收敛 $\therefore \sum \frac{1}{1+\sqrt{2}+\dots+\sqrt{n}}$ 收敛

2. 作 $S(x) = \sum_{n=1}^{\infty} nx^{n-1}$ $\int_0^x S(x) dx = \sum_{n=1}^{\infty} x^n = \frac{x}{1-x}$

$S(x) = \frac{1}{(1-x)^2}$ $\therefore \sum_{n=1}^{\infty} \frac{n}{2^n} = 2$ -----

3. $f(x) = \arcsin x$

$f'(x) = (1-x^2)^{-\frac{1}{2}} = 1 + \frac{1}{2} \cdot \frac{x^2}{1!} + \frac{1 \cdot 3}{2^2 \cdot 2!} x^4 + \dots + \frac{1 \cdot 3 \dots (2n-1)}{2^n \cdot n!} x^{2n} + \dots$

$f(x) = f(x) - f(0) = x + \frac{1}{2} \cdot \frac{1}{1!} \frac{x^3}{3} + \dots + \frac{1 \cdot 3 \dots (2n-1)}{2^n \cdot n! (2n+1)} x^{2n+1} + \dots$

$\frac{f^{(99)}(0)}{99!} = \frac{1 \cdot 3 \dots 97}{2^{49} \cdot 49! \cdot 99} \Rightarrow f^{(99)}(0) = (99!!)^2$ -----

4. $\therefore \sum a_n x^n$ 在 $x=1$ 处条件收敛

$\therefore \sum a_n x^n$ 的收敛半径 $R=1$.

($\because$ 若 $R>1$, 则 $\sum a_n x^n$ 在 $x=1$ 处绝对收敛, $\therefore R \leq 1$,
又若 $R<1$, 则 $\sum a_n x^n$ 在 $x=1$ 处发散, $\therefore R \geq 1$)

$\therefore \sum a_n x^{n+1}$ 的收敛半径 $R=1$, 据定理,

$\therefore \sum a_n (n+1) x^n$ 的收敛半径 ----- 4.5

5. 求 $f(x) = e^x$ ($0 \leq x \leq \pi$) 的周期为 2π 的傅里叶级数

解法: $\therefore 2l = 2\pi, l = \pi$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} e^x dx = \frac{e^{2\pi} - 1}{\pi}$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} e^x \cos nx dx = \frac{e^{2\pi} - 1}{\pi(n^2 + 1)} \quad (n=1, 2, \dots) \quad \dots 2\frac{1}{2}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} e^x \sin nx dx = -\frac{n(e^{2\pi} - 1)}{\pi(n^2 + 1)} \quad \dots 2\frac{1}{2}$$

$$\therefore \frac{e^{2\pi} - 1}{2\pi} + \sum_{n=1}^{+\infty} \left(\frac{e^{2\pi} - 1}{\pi(n^2 + 1)} \cos nx - \frac{n(e^{2\pi} - 1)}{1 + n^2} \sin nx \right) = \begin{cases} e^x & 0 < x < 2\pi \\ \frac{e^{2\pi} + 1}{2} & x = 0, 2\pi \end{cases}$$

解法: $a_n + bi = \frac{1}{\pi} \int_0^{2\pi} e^x (\cos nx + i \sin nx) dx = \frac{1}{\pi} \int_0^{2\pi} e^{(n+i)x} dx = \dots$ (6分) 最后2分. $\dots 2\frac{1}{2}$

6. 求原点关于平面 $\pi: 2x - 3y + 4z + 29 = 0$ 的对称点.

解 $L: \frac{x}{2} = \frac{y}{-3} = \frac{z}{4} = t, \begin{cases} x = 2t \\ y = -3t \\ z = 4t \end{cases} \quad \dots 4\frac{1}{2}$

$$29t + 29 = 0, t = -1, \text{ 对称点的 } t = -2 \quad \dots 4\frac{1}{2}$$

$\therefore$ 对称点 $P: (-4, 6, -8)$

7. 通过 $M_1(1, 1, 1)$ 和 $M_2(2, 3, 4)$, 作与平面 $x + y + z = 1$ 垂直的平面

解: 平面的法向量 $\vec{n} \perp \overrightarrow{M_1 M_2}$, $\vec{n} \perp (\vec{i} + \vec{j} + \vec{k})$

$$\text{所以 } \vec{n} = \overrightarrow{M_1 M_2} \times (\vec{i} + \vec{j} + \vec{k}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{vmatrix} = -5\vec{i} + 4\vec{j} - \vec{k}$$

$\therefore$ 所求平面方程 $-5(x-1) + 4(y-1) - (z-1) = 0$

或 $5x - 4y + z - 2 = 0$

8. 过点 $A(1, 0, 4)$, 且平行于 $3x - 4y + z = 0$, 又与直线 $\frac{x+1}{1} = \frac{y-3}{1} = \frac{z-2}{2}$ 相交的直线方程

解法(一) 过 A 且平行已知平面的平面 $3(x-1) - 4(y-0) + (z-4) = 0$

过已知直线的平面束 $\lambda(x+1-y+3) + \mu(y-3-\frac{z}{2}) = 0$ 4分

代入 $A(1, 0, 4)$, 得 $3\lambda - 4\mu = 0$, $\therefore 10x - 4y - 3z + 22 = 0$

从而所求直线 $\begin{cases} 3x - 4y + z - 1 = 0 \\ 10x - 4y - 3z + 22 = 0 \end{cases}$ 4分

解法(二): 过点 $A(1, 0, 4)$ 的直线与已知直线的交点 $B(a, a+4, 2a+2)$

$\overrightarrow{AB} = \{a+1, a+4, 2a-2\} \perp \vec{n} = \{3, -4, 1\}$ 4分

$\overrightarrow{AB} \cdot \vec{n} = 0$ 得 $a = 15$, $\therefore \overrightarrow{AB} = \{16, 19, 28\}$

所求直线: $\frac{x+1}{16} = \frac{y}{19} = \frac{z-4}{28}$ 4分

9. 求 $L: \begin{cases} x+y+z=0 \\ y-z-1=0 \end{cases}$ 绕 z 轴的旋转面方程

解: 任取 $M(x, y, z)$ 在旋转面上, 过 M 作垂直于 z 轴的平面

该平面与 L 相交于 $M_1(x_1, y_1, z_1)$

有关系 $\begin{cases} \sqrt{x_1^2 + y_1^2} = \sqrt{x^2 + y^2} \\ z_1 = z \end{cases}$ 4分

又 M_1 在 L 上: $\begin{cases} x_1 + y_1 + z_1 = 0 \\ y_1 - z_1 - 1 = 0 \end{cases}$

利用关系, 消去 x_1, y_1, z_1 得: $x^2 + y^2 = 5z^2 + 6z + 2$ 4

10. 已知 $z = (e^x + y^2)^y$ 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$

解 $\frac{\partial z}{\partial x} = y(e^x + y^2)^{y-1} e^x$... 4分 $\frac{\partial z}{\partial y} = (e^x + y^2)^y (\ln(e^x + y^2) + \frac{2y^2}{e^x + y^2})$

11. $u = f(x, \frac{x}{y})$, 求 $\frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x \partial y}$

解: $\frac{\partial u}{\partial x} = f'_1(x, \frac{x}{y}) + f'_2(x, \frac{x}{y}) \cdot \frac{1}{y}$ -----

$\frac{\partial^2 u}{\partial x \partial y} = f''_{12} \cdot (-\frac{x}{y^2}) + f''_{22} \cdot (-\frac{x}{y^3}) - \frac{1}{y^2} f'_2$ -----

12. 解: $u = f(z, \eta)$ $z = x+ay$ $\eta = x-y$

$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial z} + \frac{\partial u}{\partial \eta}$, $\frac{\partial u}{\partial y} = \frac{\partial u}{\partial z} \cdot a + \frac{\partial u}{\partial \eta} (-1)$ -----

$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial z^2} + 2 \frac{\partial^2 u}{\partial z \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}$, $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial z^2} a + \frac{\partial^2 u}{\partial z \partial \eta} (a-1) - \frac{\partial^2 u}{\partial \eta^2}$

$\frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial z^2} a^2 - 2a \frac{\partial^2 u}{\partial z \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}$ -----

代入原方程得

$(1+4a+3a^2) \frac{\partial^2 u}{\partial z^2} + (2+4(-1+a)-6a) \frac{\partial^2 u}{\partial z \partial \eta} = 0$

令 $1+4a+3a^2=0$, $(2+4(-1+a)-6a) \neq 0$ 得 $a = -\frac{1}{3}$

13. (1) $\because 0 < a_0 < 1$, 不难看出 $0 < a_n < 1$ ($n=1, 2, \dots$)

由 $a_{n+1} = a_n(1-a_n) < a_n$ $\therefore \{a_n\}$ 单调减少 ----- 2分

记 $\lim_{n \rightarrow \infty} a_n = A$, 由 $a_{n+1} = a_n(1-a_n)$, 得 $A = A(1-A)$ $\therefore A=0$

由莱布尼兹定理, 得 $\sum (-1)^n a_n$ 收敛. ----- 2分

(2) $\because 0 < a_0 < 1$, 且 $0 < a_n < a_0 < 1$ ($\forall n$)

$a_{n+1} = a_n(1-a_n)$, $\frac{1}{a_{n+1}} = \frac{1}{a_n} + \frac{1}{1-a_n}$, $\frac{1}{a_{n+1}} - \frac{1}{a_n} = \frac{1}{1-a_n} < \frac{1}{1-a_0} = b$

$\therefore \frac{1}{a_1} - \frac{1}{a_0} = \frac{1}{1-a_0} = b$, $\frac{1}{a_2} - \frac{1}{a_1} < b$, $\dots$, $\frac{1}{a_n} - \frac{1}{a_{n-1}} < b$

$\therefore \frac{1}{a_n} - \frac{1}{a_0} \leq nb$, $\frac{1}{a_n} \leq \frac{1}{a_0} + nb$ $\therefore a_n \geq \frac{1}{nb + \frac{1}{a_0}}$

$\therefore \sum a_n$ 收敛 ----- 4分